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QQ–N–288
AMENDMENT—1
September 21, 1961

FEDERAL SPECIFICATION

NICKEL-COPPER ALLOY AND NICKEL-COPPER-SILICON ALLOY CASTINGS

This amendment, which forms a part of Federal Specification QQ–N–288, dated December 20, 1956, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Page 1, paragraph 1.1. Delete and substitute:

"This specification covers six nickel copper alloys suitable for the production of sand, centrifugal and investment castings with varying percentages of silicon (see 6.5)."

Delete all reference to "monel" in the specification and substitute "nickel-copper".

Page 1, paragraph 1.2.1. Add the following:

"Composition F–Ni–Cu–Si–C (Investment Casting)."

Page 1, paragraph 1.2.2. Delete and substitute:

1.2.2 *Condition.*—Unless otherwise specified in the contract or order, castings shall be furnished in the "as cast".

Page 2, paragraph 3.1.2. Delete and substitute:

3.1.2 An analysis of each melt (or master heat in the case of investment castings) shall be furnished by the contractor, showing the percentage of the elements designated in table I.

Page 2, following paragraph 3.1.2.1 add new paragraphs 3.1.2.2 and 3.1.2.2.1:

3.1.2.2 The metal used shall be of a quality suitable to produce castings which comply with all the requirements of this specification. Investment castings shall be poured from either remelted heat metal or directly from a master heat.

3.1.2.2.1 A master heat used for the production of investment castings is previously refined metal of a single furnace charge. Gates, sprues, risers and the rejected castings shall not be remelted directly for pouring of castings without refining. They shall be used only for additions to the melt in the preparation of a master heat. Metal in the form of shot from more than one master heat may be uniformly blended together to form a master heat lot; the total weight of metal in a master heat shall not exceed 7000 pounds.

Composition	Nickel[4]	Copper	Silicon	Iron, maximum	Manganese, maximum	Carbon[5] maximum	Aluminum, maximum[4]	Columbium and Tantalum	Cobalt[4] maximum
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*
F	Remainder	29-34	2.3-3.0	2.5	1.5	0.40-0.70	...	...	1.0

[4] Analysis for these elements is not normally made.
[5] Except where a range is shown.

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Page 3. Add as new paragraph:

3.4.1 *Preproduction sample (investment casting only).*—When a preproduction sample is required (see 6.2), it shall meet the requirements of this specification.

Page 3, table II. Add the following:

Composition	Tensile strength, minimum	Yield strength, minimum[1]	Elongation in 2 inches, minimum
	P.s.i.	*P.s.i.*	*Percent*
F[4]	70,000	40,000	10

[4] As cast.

Page 3, paragraph 3.8, line 2. Add "and F" between "D" and "castings".

Page 4, paragraph 4.1. Delete and substitute:

4.1 Lot.—For the purpose of sampling, except for chemical analysis of investment castings, a lot shall consist of all castings produced for one pouring of a furnace, crucible, or a ladle. When two or more furnace or crucible melts are used to change a ladle for pouring, the castings produced therefrom shall comprise the lot. In the case of investment castings, for the purpose of sampling for chemical analysis, a lot shall consist of all the castings produced from one master heat.

Page 6, paragraph 4.4.2, line 3. Add "and F" between "D" and "castings".

Page 7. Add the following new paragraphs:

4.5 Bend test (applicable to investment castings only).—

4.5.1 *Test specimen.*—The test specimen shall be at least 0.090 inch in diameter or thickness and 2 inches in length.

4.5.2 *Test method.*—Bend through an angle of 30 degrees around a radius of curvature not greater than 0.5-inch in accordance with method 231 of Federal Test Method Standard No. 151.

4.6 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Page 10. Following paragraph 6.1.5 add new paragraph 6.1.6:

6.1.6 The nickel-copper-silicon alloy F is intended for applications involving high resistance to corrosion, wear and galling, non-magnetic characteristics and relative ease of machineability. Normally supplied in the "as cast" condition, it may be further softened by solution annealing where maximum machineability is required. Parts in this condition will normally be age hardened after machining. The solution annealing and age hardening procedures are the same as those shown in 6.1.4.1.

Page 10, paragraph 6.2. Add the following:

"(k) Whether a preproduction sample is required (see 3.4.1)."

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